The equivalent enclosure size (*EES*) is determined using the equivalent width and height using Equation (13) as follows:

$$EES = \frac{\text{Height}_1 + \text{Width}_1}{2} \tag{13}$$

where

Height_1	is the equivalent enclosure height
Weight_1	is the equivalent enclosure width
EES	is the equivalent enclosure size

4.8.4 Determination of enclosure size correction factor (*CF*)

The correction factor (*CF*) for a "Typical Enclosure" is obtained by using Equation (14) as follows:

$$CF = b1 \times EES^2 + b2 \times EES + b3 \tag{14}$$

Use Equation (15) for correction factor for a "Shallow Enclosure" as follows:

$$CF = \frac{1}{b1 \times EES^2 + b2 \times EES + b3} \tag{15}$$

where

$b1$ to $b3$	are the coefficients for Equation (14) and Equation (15) provided in Table 7
CF	is the enclosure size correction factor used in Equation (3) through Equation (10)
EES	is the equivalent enclosure size used to find the correction factor determined using Equation (13). For typical box enclosures the minimum value of *EES* is 20

Table 7 provides the coefficients $b1$ to $b3$ for both typical and shallow enclosure types.

Table 7—Coefficients for Equation (14) and Equation (15)

Box type	E.C.	*b*1	*b*2	*b*3
Typical	**VCB**	−0.000302	0.03441	0.4325
	VCBB	−0.0002976	0.032	0.479
	HCB	−0.0001923	0.01935	0.6899
Shallow	**VCB**	0.002222	−0.02556	0.6222
	VCBB	−0.002778	0.1194	−0.2778
	HCB	−0.0005556	0.03722	0.4778

4.9 Determination of I_{arc}, E, and AFB (600 V < V_{oc} ≤ 15 000 V)

To determine the final arcing current, incident energy, and arc-flash boundary at a specific voltage, first calculate the intermediate values for the three voltage levels of 600 V, 2700 V, and 14 300 V. Then use the interpolation Equation (16) to Equation (24) to determine the final estimated values as follows:

Arcing current (I_{arc})

$$I_{arc\_1} = \frac{I_{arc\_2700} - I_{arc\_600}}{2.1}\left(V_{oc} - 2.7\right) + I_{arc\_2700} \tag{16}$$

$$I_{arc\_2} = \frac{I_{arc\_14300} - I_{arc\_2700}}{11.6}\left(V_{oc} - 14.3\right) + I_{arc\_14300} \tag{17}$$

$$I_{arc\_3} = \frac{I_{arc\_1}\left(2.7 - V_{oc}\right)}{2.1} + \frac{I_{arc\_2}\left(V_{oc} - 0.6\right)}{2.1} \tag{18}$$

where

I_{arc_1}	is the first I_{arc} interpolation term between 600 V and 2700 V (kA)
I_{arc_2}	is the second I_{arc} interpolation term used when V_{oc} is greater than 2700 V (kA)
I_{arc_3}	is the third I_{arc} interpolation term used when V_{oc} is less than 2700 V (kA)
V_{oc}	is the open-circuit voltage (system voltage) (kV)

When $0.600 < V_{oc} \leq 2.7$, the final value of arcing current is given as follows:

$$I_{arc} = I_{arc\_3}$$

When $V_{oc} > 2.7$, the final value of arcing current is given as follows:

$$I_{arc} = I_{arc\_2}$$

The arc duration can be determined using I_{arc}. This time is used to determine the incident energy and arc-flash boundary.

Incident energy (E)

$$E_1 = \frac{E_{2700} - E_{600}}{2.1}\left(V_{oc} - 2.7\right) + E_{2700} \tag{19}$$

$$E_2 = \frac{E_{14300} - E_{2700}}{11.6}\left(V_{oc} - 14.3\right) + E_{14300} \tag{20}$$

$$E_3 = \frac{E_1\left(2.7 - V_{oc}\right)}{2.1} + \frac{E_2\left(V_{oc} - 0.6\right)}{2.1} \tag{21}$$

where

E_1	is the first E interpolation term between 600 V and 2700 V (J/cm^2)
E_2	is the second E interpolation term used when V_{oc} is greater than 2700 V (J/cm^2)
E_3	is the third E interpolation term used when V_{oc} is less than 2700 V (J/cm^2)

Arc-flash boundary (*AFB*)

$$AFB_1 = \frac{AFB_{2700} - AFB_{600}}{2.1}\left(V_{oc} - 2.7\right) + AFB_{2700} \tag{22}$$

$$AFB_2 = \frac{AFB_{14300} - AFB_{2700}}{11.6}\left(V_{oc} - 14.3\right) + AFB_{14300} \tag{23}$$

$$AFB_3 = \frac{AFB_1\left(2.7 - V_{oc}\right)}{2.1} + \frac{AFB_2\left(V_{oc} - 0.6\right)}{2.1} \tag{24}$$

where

AFB_1 is the first *AFB* interpolation term between 600 V and 2700 V (mm)
AFB_2 is the second *AFB* interpolation term used when V_{oc} is greater than 2700 V (mm)
AFB_3 is the third *AFB* interpolation term used when V_{oc} is less than 2700 V (mm)

When $600 < V_{oc} \leq 2.7$, the final values of incident energy and arc-flash boundary are given as follows:

$$E = E_3$$

$$AFB = AFB_3$$

When $V_{oc} > 2.7$, the final values of incident energy and arc-flash boundary are given as follows:

$$E = E_2$$

$$AFB = AFB_2$$

It is recommended to calculate a second set of arc duration, incident energy, and arc-flash boundary values based on the reduced arcing current I_{arc_min} to account for the arcing current variation effect on the operation of protective devices. The final incident energy or arc-flash boundary is the higher of the two calculated values.

The incident energy (cal/cm²) is obtained by dividing E by 4.184 (1 cal = 4.184 J). See B.2.

4.10 Determination of I_{arc}, E, and AFB ($V_{oc} \leq 600$ V)

This subclause describes how to determine the final arcing current, incident energy, and arc-flash boundary for a specific open-circuit voltage, 208 V ≤ V_{oc} ≤ 600 V. First, calculate the arcing current using Equation (25). Using the arcing current, estimate the arc duration and proceed to determine the incident energy and arc-flash boundary.

Arcing current (I_{arc})

The final arcing current can be determined using Equation (25).

$$I_{arc} = \frac{1}{\sqrt{\left[\frac{0.6}{V_{oc}}\right]^2 \times \left[\frac{1}{I_{arc\_600}{}^2} - \left(\frac{0.6^2 - V_{oc}{}^2}{0.6^2 \times I_{bf}{}^2}\right)\right]}} \tag{25}$$

where

- V_{oc} is the open-circuit voltage (kV)
- I_{bf} is the bolted fault current for three-phase faults (symmetrical rms) (kA)
- I_{arc} is the final rms arcing current at the specified V_{oc} (kA)
- I_{arc_600} is the rms arcing current at V_{oc} = 600 V found using Equation (1) (kA)

The arc duration can be determined using I_{arc}. This time is used to determine the incident energy and arc-flash boundary.

Incident energy (E)

The incident energy is given as follows:

$$E = E_{\leq 600}$$

where

- $E_{\leq 600}$ is the incident energy for V_{oc} ≤ 600 V determined using Equation (6) solved using the arc current determined from Equation (1) and Equation (25) (J/cm²)
- E is the final incident energy at specified V_{oc} (J/cm²)

Arc-flash boundary (AFB)

The arc-flash boundary is given as follows:

$$AFB = AFB_{\leq 600}$$

where

- $AFB_{\leq 600}$ is arc-flash boundary for V_{oc} ≤ 600 V determined using Equation (10) solved using the arc current determined from Equation (1) and Equation (25) (mm)
- AFB is the final arc-flash boundary at specified V_{oc} (mm)

Calculate a second set of arc duration, incident energy, and arc-flash boundary values based on the reduced arcing current I_{arc_min} to account for the arcing current variation effect on the operation of protective devices. The final incident energy or arc-flash boundary is the higher of the two calculated values.

4.11 Single-phase systems

This model does not cover single-phase systems. Arc-flash incident energy testing for single-phase systems has not been researched with enough detail to determine a method for estimating the incident energy. Single-phase systems can be analyzed by using the single-phase bolted fault current to determine the single-phase arcing current (using the equations provided in 4.4 and 4.10). The voltage of the single-phase system (line-to-line, line-to-ground, center tap voltage, etc.) can be used to determine the arcing current. The arcing current can then be used to find the protective device opening time and incident energy by using the three-phase equations provided in this guide. The incident energy result is expected to be conservative.

4.12 DC systems

Arc-flash incident energy calculation for dc systems is not part of this model. However, publication references (Ammerman et al. [B1], Das [B16], [B17], Doan [B25], Klement [B62]) provide some guidance for incident energy calculation.

5. Applying the model

The purpose of this clause is to provide an overview of the analysis process required to apply the model. The steps described in this clause may be applied manually, but it may be more convenient to use available short-circuit and protective device coordination programs, which have embedded the steps necessary to apply the model.

Clause 6 provides the following summary of considerations and steps necessary to apply the calculation model:

— An overview of system data collection requirements is provided in 6.2. Accurate data collection is an important part of the study process.

— Calculation of bolted fault current levels, considering system operating modes, is discussed in 6.3 and 6.4.

— Information on equipment-related parameters that are used in the model, such as equipment dimensions, electrode configuration, and working distance, are discussed in 6.5, 6.6, and 6.7.

— Subclauses 6.8 and 6.9 discuss calculation of the arcing current and determination of the arcing duration to be used in the model.

— Subclauses 6.10 and 6.11 address the calculation of the final incident energy and arc-flash boundary. The discussion of specific equipment types in Annex C may also be useful.

To illustrate the model application process, two detailed calculation examples are provided in Annex D.

NOTE—Subclause 4.3 covers the equation application procedure.

6. Analysis process

6.1 General overview

An arc-flash hazard analysis can be performed in association with or as a continuation of a short-circuit study and protective-device coordination study. A complete coordination study may not be required, but the protective device opening time in response to arcing currents must be applied during the analysis process. The process and methodology of calculating short-circuit currents and performing protective-device coordination is covered in standards such as IEEE Std 551™ (*IEEE Violet Book*™),[10] IEC 60909-0 [B51],[11] and IEEE Std 242™ (*IEEE Buff Book*™). The results of the short-circuit study enable calculation of the arcing fault currents at selected locations. Protective device time response to the arcing currents is used to evaluate the time required for the protective devices to interrupt during fault conditions.

Deliverables of the arc-flash hazard analysis calculation are the arc-flash boundary and the arc-flash incident energy at defined working distances from the arcing source at the selected locations in the electrical system. The results of the study document the incident energy analysis and may be used by workers as part of an overall electrical safety risk assessment.

6.2 Step 1: Collect the system and installation data

A significant effort in performing an arc-flash hazard study is the collection of electrical system data. Even for a facility with nominally up-to-date single-line diagrams, time-current curves, and short-circuit model on a computer, the data collection portion of the study may take about half of the effort. Even for new facilities, field verification of the single-line diagrams and protection settings is necessary to verify the integrity of documentation of the power system. Facility workers who are familiar with the electrical system and its safety-related work practices may be able to assist or perform this part of the study. Refer to IEEE Std 1584.1 for further information on system data required for an arc-flash hazard analysis.

While the data required for this study is similar to data collected for typical short-circuit and protective-device coordination studies, it goes further in that all low-voltage distribution and control equipment within the scope of the study up through its sources of supply must be included.

Collect information to perform incident energy calculations on electrical equipment that is likely to require examination, adjustment, servicing, or maintenance while energized. This could include equipment such as low- and medium-voltage switchgear, medium-voltage plug-in connectors, motor starters, motor control centers (MCCs), switchboards, switchracks, panelboards, separately-mounted switches and circuit breakers, ac and dc drives, power distribution units (PDUs), uninterruptible power supplies (UPS), transfer switches, industrial control panels, meter socket enclosures, etc.

The study process begins with a review of available single-line diagrams and electrical equipment site and layout arrangement with people who are familiar with the site. The diagrams should be updated to show the current system configuration.

Electrical system studies should have an up-to-date single-line diagram(s). The single-line diagrams include all alternate feeds.

Follow applicable industry standards for performing short-circuit studies. See Clause 2 for examples of industry standards.

Obtain the available fault current and X/R ratio that represents the source. For transformers, generators, large motors, and switchgear, collect relevant nameplate data such as voltage/voltage ranges or tap settings, ampacity, kilowatt or kilovoltamperes, first cycle (momentary or close and latch) and/or interrupting current rating, impedance or transient/subtransient reactance data. Because information regarding box (enclosure) size and electrode configuration may be needed for more detailed calculations, it may be necessary to take measurements if possible or collect other data such as nameplate information or device catalog numbers that will allow for relevant equipment enclosure dimensions and configurations to be estimated.

Next, collect conductor and cable data along with its installation (routing and support method, in magnetic raceway-steel conduit or non-magnetic raceway-aluminum conduit, etc.) for all electrical circuits between the

[10] Information on references can be found in Clause 2.
[11] The numbers in brackets correspond to those of the bibliography in Annex A.